

What Is Math Talk & How Can We Use It to Support Numeracy Learning?

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Road Map for the Day

1. What is math talk? – Insights from basic research but also classroom observations
2. Productive math talk: I show you, we try it together, then you try it
3. Math talk beyond numeracy lessons

Whose talk?

What's the purpose of talk?

What is math talk?

- Talk between adults and children about math, part of everyday classroom moments (numeracy lessons and beyond).
- For children: to reason out loud, to practice talking about numbers and math
- For adults: to discover, check, extend, model
- The point is the **process**, not the product.
- Through talking, we're working out our ideas, finding out what each party is thinking or asking

Why it matters

Developmental Psychology
2006, Vol. 42, No. 1, 59–69

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Preschool Children's Mathematical Knowledge: The Effect of Teacher "Math Talk"

Raquel S. Klibanoff, Susan C. Levine, and
Janellen Huttenlocher
University of Chicago

Larry V. Hedges
University of Chicago



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Journal of Experimental Child
Psychology

journal homepage: www.elsevier.com/locate/jecp



Assessing the association between parents' math
talk and children's math performance: A
preregistered meta-analysis

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Two notions of “math talk”

What you talk about

Developmental studies (Levine et al., 2010; Gunderson et al., 2011)

- Count objects together.
- More/Less, naming quantities, simple sums
- More math language → stronger early number knowledge.

How you respond

Classroom discourse studies (Ruede et al., 2003)

- Questions that open up a dialogue and wait time.
- Makes thinking explicit and extends children’s thinking

Two follow-ups, where do they lead?

- One child response.
- Two possible teacher follow-ups.
- What's the difference? Where does each conversation lead?

One response, two follow-ups

Setting: a child is sorting crayons, 4 red crayons and 6 blue ones. Child was making a spontaneous comment

Child: “There are more blue ones.”

T: “Yes, you’re right”

T: “How did you know?
Can you show me?”

What does the second teacher learn about the child that the first teacher doesn’t?

Some follow-ups end the conversation earlier than others.

Productive Math Talk in the Classroom: Example 1

*Setting: on the table, two rows of buttons. Row A has 5 and row B has 7. **B has more***

T: Which row has more buttons?

C: A.

T: How do you know?

C: Because there's more.

T: Can you show me?

C: [counts A] 1-2-3-4-5. [counts B] 1-2-3-4-5-6-7

T: So which has more buttons?

C: B!

T: How do you know?

C: I counted! B has 7

T: That's right, B has more buttons, because 7 is more than 5. And you counted to find out, that was good.

Alternatively:

T: Which row has more buttons?

C: A.

T: Are you sure? Count again.

C: [counts A] 1-2-3-4-5. [counts B] 1-2-3-4-5-6-7

T: Good, so B has more.

What's the difference between these two?

What's the smallest change between these two cases that make the biggest difference?

Telling vs. asking

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T: Good, so B has more.

Productive moves: asking children to demonstrate without giving away how to solve the problem, modelled math talk at the end, highlighted the reasoning more explicitly.

Productive Math Talk in the Classroom: Example 2

*Setting: T started with having a ball and asked kids to find an object smaller than it. Children compare a ball with a blue cube, and **the cube is smaller**.*

T: [Child 0] picked the blue cube. When I put it beside the ball, which is smaller, the ball or the cube?

C1: The cube.

T: The cube. [Child 2], what do you think?

C2: The ball.

T: Why do you say the ball is smaller?

C2: You look at the size.

T: Can you show me? Put them side by side.

C2: [compares them] Oh... the cube.

T: The cube is smaller. How do we know?

C2: Because the ball is bigger. There's more here.

T: Can someone repeat what [Child 2] said?

C3: The ball is bigger because it's longer there.

.... **Another child (C3) picked up another object**

T: Does everyone agree? Show me how you can tell.

Alternatively:

T: [Child 0] picked the blue cube. That's right.

Productive moves: polling multiple children, getting children to implement a procedure themselves to find out, "why" and "show me"

Outcome: a child revises her own thinking

3 moves to keep in your pocket

1. Explain

*Help children share and expand their own ideas.
Let you see where their ideas are. Help them
reason*

- “Can you say more about that?”
- “Why do you think that?”
“Can you show me?”

2. Listen

Help children listen to each other.

- “What did [XXX] say?”
- “Who can repeat what she said?”

3. Think with others

Help children build on each other.

- “Who can add to that?”
- “Do you agree or disagree?”

3 moves to keep in your pocket

1. Explain

*Help children share and expand their own ideas.
Let you see where their ideas are. Help them
reason*

- “Can **you** say more about that?”
- “Why do **you** think that?”
“Can **you** show me?”

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Help children listen to each other.

- “What did [XXX] say?”
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Help children build on each other.

- “Who can add to that?”
- “Do you agree or disagree?”

Productive Math Talk in the Classroom: Example 1

*Setting: on the table, two rows of buttons. Row A has 5 and row B has 7. **B has more***

T: Which row has more buttons?

C: A.

T: How do you know? ← Explain

C: Because there's more.

T: Can you show me? ← Explain

C: [counts A] 1-2-3-4-5. [counts B] 1-2-3-4-5-6-7

T: So which has more buttons?

C: B!

T: How do you know? ← Explain

C: I counted! B has 7

T: That's right, B has more buttons, because 7 is more than 5. And you counted to find out, that was good. ← Modelling

Productive moves: asking children to demonstrate without giving away how to solve the problem, modelled math talk at the end, highlighted the reasoning more explicitly.

Productive Math Talk in the Classroom: Example 2

*Setting: Children compare a ball with a blue cube, and **the cube is smaller**.*

T: [Child 0] picked the blue cube. When I put it beside the ball, which is smaller, the ball or the cube?

C1: The cube.

T: The cube. [Child 2], what do you think?

Invite others

C2: The ball.

T: Why do you say the ball is smaller?

Explain

C2: You look at the size.

T: Can you show me? (Put them side by side.)

Explain

C2: [compares them] Oh... the cube.

T: The cube is smaller. How do we know?

Explain

C2: Because the ball is bigger. Look, there's more here

T: Can someone repeat what [Child 2] said?

Listen

C3: The ball is bigger because it's longer there.

.... Another child (C3) picked up another object

T: Does everyone agree? Show me how you can tell.

Invite others

Why it matters

- Makes children's thinking visible to them and to us.
- Builds mathematical reasoning, along with procedure.
- Reveals what they actually understand (or do not understand), not just what they can do.
- Sets a culture that it's okay to make mistakes! We can talk about them.
- Sets a culture that it's not only the product that matters

Math talk with mistakes

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T: Good, so B has more.

Math talk with mistakes

- Errors are learning opportunities! Not just things to correct and move on.
 - A wrong answer is a window into how the child is thinking.
 - How we react to mistakes shapes whether children are willing to try things out
 - We can practice to respond with **curiosity** and not judgment.
- The 3 productive math talk moves

Responding to a mistake

- Pause before the impulse to correct/provide answer kicks in
- Is there anything you can find out about the child's thinking?
- **BE NEUTRAL!**
- Ask to explain: "How do you know?" or "How did you get that?"
- Ask to show: "Can you show me?"

Provide answer or model the procedure if you think child is lost, is not following, or is silent.

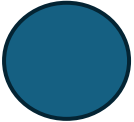
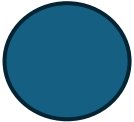
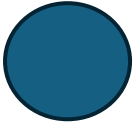
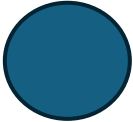
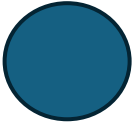
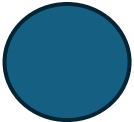
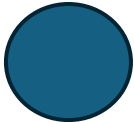
Part 2: We try + You try

Demo: math talk during numeracy lessons

15 minutes

We try – Counting

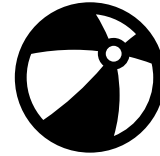
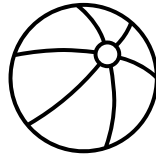
Make 6



We try – Patterning

Make ABAB pattern



“You try”

- 6 groups of 5 people + 2 groups of 4.
- Each group gets a scenario.
- Take turns being the teacher and the child — role-play the conversation. **(WRITE IT DOWN)**
- Each scenario involves a math moment, a mistake, or an unexpected response.
- Share within your table, then volunteer to share with the broader group

Aisha's blocks

Aisha is working on a learning basket activity on her own. There are digit cards (3, 4, 5, 6) and she has to match each digit with the corresponding number of blocks.

Aisha has just finished when the teacher comes over. The 3 and 4 are matched correctly. For the 5, she's put 6 blocks; for 6, she's put 8.

The teacher pauses. What can she do next?

What is the child thinking? What might she actually know, or be confused about?

Strawberries and watermelons

The class is shown two pictures: one with 5 small strawberries, one with 3 large watermelons.

The teacher asks which picture has more fruits.

Sally says: “Watermelon!”

What can the teacher do next?

What is the child thinking? What might she actually know, or be confused about?

Happy and sad

Each child has two cards, happy or sad face. During a large-group activity, children show either a happy or sad face card when the music stops. The teacher sorts them into two groups, “happy” and “sad”, and counts each group to decide which has more.

What's the teaching moment here?
What would you do?

Number cards

Each child picks a digit card and takes that many counters from a basket. The teacher walks around checking each child's pile. When a child has the wrong number, she tends to say things like:

“Why do you have extra?”

“Why are there so many?”

“How come you have 4? Can you count again?”

What do they assume before the child has spoken?

Is this the same "why" we use to invite reasoning?

What would you ask instead?

Top and bottom

Children are told to paint the top picture black and the bottom picture red. One child does the opposite.

The teacher came over and said, “This one red, this one black,” and asked the child to correct it.

What might the child have been thinking?

What does the child learn from the teacher’s response?

What would you say?

Reflecting on the activity

- What was the first thing you wanted to say?
- And what did you say instead?
- Where did the conversation open up? Where did it stall?
- Can you compare this to how you may have done it in the past? What have you learned?

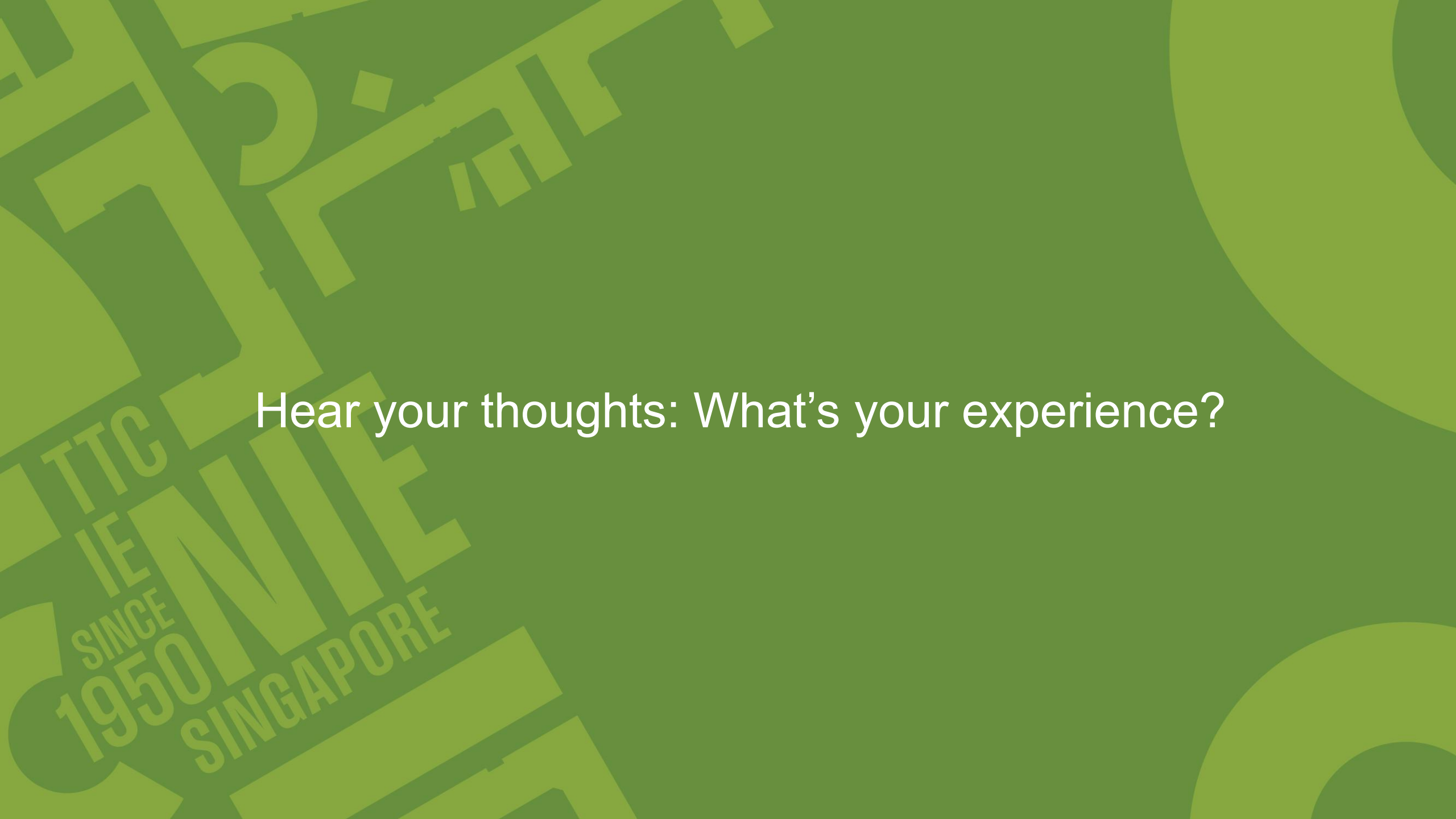
Part 3

Teachable math moments beyond numeracy lessons

5 minutes

Math talk beyond numeracy lessons

- Transitions, snack, outdoor play, free play – EVERY routine has math in it!!
- Quantity, comparison, shape, position, sequence, measurement
- Once we practice “opening up our eyes to these math moments”, they’ll become easier to spot and you can use them to engage children with numeracy concepts.

The background is a solid green color. On the left side, there is a faint, stylized graphic of a building or structure with the text 'TTC' and 'SINCE 1950 SINGAPORE' integrated into it. On the right side, there are large, overlapping circles in a lighter shade of green.

Hear your thoughts: What's your experience?

Straightforward ones

- Counting children when lining up for outdoor walk
- How many children came to school today?
- Showing numbers with fingers (esp for N1, N2 children)
- Asking “how many” in general, across different settings

Less common ones

- Going to the park, spatial talk (turn left, turn right, at this corner, go straight, etc)
- Hunting for shapes on your outdoor walk
- Hunting for patterns
- Who's going to finish lunch first? Second? Third? ... Second last? Last?
- Block play: "Let's build a tower then destroy it!" "I'm going to build a really tall tower"

Small group activity

- Each group gets a bag of objects, a picture, or a routine/outdoor scenario.
- Discussion Q1: What math questions might you ask?
- Discussion Q2: What question may you ask of an N1 or N2 children?
Or a question that you may ask of K1 or K2 children?
- Discussion Q3: Can you think of an unexpected response? How might you respond to it?

Share and Reflection

- One question that your group really liked
- One “unexpected child response” and what you'd do with it.
- One thing you'd take into Monday morning.

Two notions of “math talk”

What you talk about

Developmental studies (Levine et al., 2010; Gunderson et al., 2011)

- Count objects together.
- More/Less, naming quantities, simple sums
- More math language → stronger early number knowledge.

How you respond

Classroom discourse studies (Ruede et al., 2003)

- Questions that open up a dialogue and wait time.
- Makes thinking explicit and extends children’s thinking

Take-home

- **Prioritize** what you want to practice. Some examples below:
 - Talking about math outside of numeracy lessons
 - Having ONE longer conversation about math next time a child asks you a math-relevant question
 - Noticing mistakes and thinking about “oh what can I say here?”
 - Following up on mistakes and trying out the strategies
- **Awareness** is key. That’s the first step. Strategies come next.
- Try something do-able FOR YOU.

Thank you

Questions?

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